Convergence Study on Polycrystalline Materials

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Polycrystalline materials possess distinct material properties at the individual grain scale, creating jumps (discontinuities) in properties at grain boundaries. This leads to limitations in solution regularity and global convergence rates of finite element calculations. This study examines convergence rates on a grain-by-grain basis, focusing on the convergence of grain-averaged quantities. We utilize conductive heat transfer and linear elasticity as model problems to investigate the relationship between relative errors and grain size.